

Given:
Due:

Further Maths (Statistics)

HW 23

Name:

Chapter 6: Chi-Squared tests (goodness of fit)

Q1. A spinner used for a game is designed to give scores with the following probabilities

The spinner is spun 80 times and the results are as follows

Test, at the 10% level of significance, whether or not the spinner is giving scores as it is designed to do. Show your working and state your hypotheses clearly.

(Total for question = 7 marks)

Score	1	2	3	4	6
Probability	$\frac{3}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{2}{5}$	$\frac{1}{10}$

Score	1	2	3	4	6
Frequency	15	4	12	41	8

This question was generally answered very well. Some lost marks for the hypotheses: statements such as the "observed frequencies match the expected frequencies" or "a uniform distribution is a suitable model" were given rather than a comment about the spinner's design such as "the spinner is giving probabilities as it is designed to do". The expected frequencies were usually found correctly but a number of candidates lost marks for incorrectly combining the scores for 1 and 2 because the observed frequency for a score of 2 was less than 5; candidates need to understand that the amalgamation of cells is only required when expected frequencies fall below 5. The calculations and selection of the critical value were usually correct and most realised that the test suggested the spinner was not working as designed.

Scheme							Marks	AO
H_0 : Spinner is working as designed (o.e.) H_1 : Spinner is not working as designed (o.e.)							B1	1.2
E_i	24	8	8	32	8		M1	3.4
O_i	15	4	12	41	8		A1	1.1b
$\frac{(O_i - E_i)^2}{E_i}$	$\frac{81}{24}$	$\frac{16}{8}$	$\frac{16}{8}$	$\frac{81}{32}$	0		M1	1.1b
$\frac{O_i^2}{E_i}$	$\frac{225}{24}$	$\frac{16}{8}$	$\frac{144}{8}$	$\frac{1681}{32}$	$\frac{64}{8}$			
$\sum \frac{(O_i - E_i)^2}{E_i} = 3.375 + 2 + 2 + 2.53125 + 0 = 9.90625$							A1	1.1b
or $\sum \frac{O_i^2}{E_i} - N = 9.375 + 2 + 18 + 52.53125 + 8 - 80 = 9.90625$								
$\nu = 5 - 1 = 4$ so $\chi^2_{4}(10\%)$ cv = 7.779 or better							B1	3.4
Result is significant so there is evidence that the spinner is not operating as designed							A1cso	3.5a
							(7 marks)	
Notes								
1 st B1 for both hypotheses given in suitable context 1 st M1 for using the model to find at least 2 correct expected frequencies 1 st A1 for all correct E_i 2 nd M1 for attempt to find test statistic (at least two correct expressions, fractions or decimals) 2 nd A1 for a correct test statistic (awrt 9.91) [accept $\frac{317}{32}$] 2 nd B1 for correct critical value (allow 7.78) NB p -value = 0.042036...so allow awrt 0.042 3 rd A1cso dep on <u>all</u> previous marks for a correct conclusion in context (can be in terms of model or spinner's design) Must mention spinner and scores <u>or</u> design Accept "spinner is not accurate"								

Q2. The number of hurricanes per year in a particular region was recorded over 80 years. The results are summarised in Table 1 below.

No of hurricanes, h	0	1	2	3	4	5	6	7
Frequency	0	2	5	17	20	12	12	12

Table 1

(a) Write down two assumptions that will support modelling the number of hurricanes per year by a Poisson distribution.

(2)

(b) Show that the mean number of hurricanes per year from Table 1 is 4.4875

(2)

(c) Use the answer in part (b) to calculate the expected frequencies r and s given in Table 2 below to 2 decimal places.

h	0	1	2	3	4	5	6	7 or more
Expected frequency	0.90	4.04	r	13.55	s	13.65	10.21	13.39

(3)

Table 2

(d) Test, at the 5% level of significance, whether or not the data can be modelled by a Poisson distribution. State your hypotheses clearly.

(6)

(Total 13 marks)

Many lost marks in part (a) by not mentioning hurricanes, as did many in part (b) who failed to show enough working. Even some of the better candidates carelessly lost marks here. The calculations for r and s were well done, as was the calculation of the test statistic. However many candidates lost a mark by including the value of the mean in a hypothesis. When this is acceptable is not fully understood even by some of the best candidates. There were very few who failed to pool their data and these candidates lost marks appropriately.

Question Number	Scheme EXAM PAPERS PRACTICE	Marks																																				
(a)	Hurricanes: occur singly / are independent or occur at random /are a rare event / at a constant rate	B1B1 (2)																																				
(b)	From data $\frac{1 \times 2 + 2 \times 5 + 3 \times 17 + .. + 7 \times 12}{80} = 4.4875$	M1A1 (2)																																				
(c)	<table><tr><td>No of hurricanes, h</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7+</td></tr><tr><td>80P($X = h$)</td><td>0.9</td><td>4.038</td><td>9.06...</td><td>13.55</td><td>15.205</td><td>13.647...</td><td>10.206...</td><td>13.388...</td></tr><tr><td>Combine to give expected frequencies > 5</td><td colspan="3">13.9991</td><td>13.55</td><td>15.205...</td><td>13.647...</td><td>10.206...</td><td>13.388...</td></tr><tr><td>Observed</td><td colspan="3">7</td><td>17</td><td>20</td><td>12</td><td>12</td><td>12</td></tr></table>	No of hurricanes, h	0	1	2	3	4	5	6	7+	80P($X = h$)	0.9	4.038	9.06...	13.55	15.205	13.647...	10.206...	13.388...	Combine to give expected frequencies > 5	13.9991			13.55	15.205...	13.647...	10.206...	13.388...	Observed	7			17	20	12	12	12	M1A1A1 (3)
No of hurricanes, h	0	1	2	3	4	5	6	7+																														
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Observed	7			17	20	12	12	12																														
(d)	<table><tr><td>$\frac{(O - E)^2}{E}$</td><td colspan="3">3.499...</td><td>0.876...</td><td>1.511...</td><td>0.198...</td><td>0.315...</td><td>0.143...</td></tr><tr><td>$\frac{O_i^2}{E_i}$</td><td colspan="3">3.500...</td><td>21.322...</td><td>26.306...</td><td>10.551...</td><td>14.108...</td><td>10.755...</td></tr></table> H_0: Poisson distribution is a good fit o.e. H_1: Poisson distribution is not a good fit o.e. $\sum \frac{(O_i - E_i)^2}{E_i} = 6.545..$ or $\frac{O_i^2}{E_i} = 86.545 - 80 = 6.545..$ (awrt 6.55 or 6.54) $\nu = 6 - 2 = 4$ cv is 9.488 $\chi^2_{\nu}(0.05)$ $6.545 < 9.488$ so insufficient evidence to reject H_0 (Hurricanes) can be modelled by a Poisson distribution	$\frac{(O - E)^2}{E}$	3.499...			0.876...	1.511...	0.198...	0.315...	0.143...	$\frac{O_i^2}{E_i}$	3.500...			21.322...	26.306...	10.551...	14.108...	10.755...	M1 B1 A1 B1 B1ft A1 (6) 13																		
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Notes (b) M for at least 2 terms on numerator. 359/80 only award M0A0 (c) M for 80xPoisson probability with 4.4875 and either 2 or 4. 1st A1 for awrt 9.06 and 2nd A1 for awrt 15.20 or 15.21 (d) 1st M1 for some pooling and attempting $\frac{(O - E)^2}{E}$ or $\frac{O^2}{E}$, at least 3 correct expressions or values. 1st B1 no value for parameter permitted 2nd A1 for a correct comment suggesting that Poisson model is suitable. No ft																																						



Q3. A quality control manager regularly samples 20 items from a production line and records the number of defective items x . The results of 100 such samples are given in Table 1 below.

x	0	1	2	3	4	5	6	7 or more
Frequency	17	31	19	14	9	7	3	0

Table 1

(a) Estimate the proportion of defective items from the production line.

(2)

The manager claimed that the number of defective items in a sample of 20 can be modelled by a binomial distribution. He used the answer in part (a) to calculate the expected frequencies given in Table 2.

x	0	1	2	3	4	5	6	7 or more
Expected frequency	12.2	27.0	r	19.0	s	3.2	0.9	0.2

Table 2

(b) Find the value of r and the value of s giving your answers to 1 decimal place.

(3)

(c) Stating your hypotheses clearly, use a 5% level of significance to test the manager's claim.

(7)

(d) Explain what the analysis in part (c) tells the manager about the occurrence of defective items from this production line.

(1)

(Total 13 marks)

Although most candidates answered part (a) correctly a surprising number failed to do so some not appreciating the difference between the mean number of defective items and the proportion. Part (b) was usually answered well with only a few using n as 100 instead of 20. Most candidates knew about combining classes in part (c) but they didn't always calculate the degrees of freedom correctly and this sometimes led to them failing to reject the binomial model. There were many good responses to part (d) with candidates showing a sound understanding of the implications of rejecting the binomial model.

Question number	Scheme	Marks																								
(a)	$\frac{0 \times 17 + 1 \times 31 + \dots}{17 + 31 + \dots} = \left(\frac{200}{100} = 2 \right), \quad \hat{p} = \frac{2}{20} = 0.1$ (Accept $\frac{2}{20}$ or 2 per 20)	M1, A1 (2)																								
(b)	e.g. $r = 100 \times \binom{20}{2} (0.1)^2 (0.9)^{18}$	M1																								
(c)	$r = 28.5, s = \text{AWRT } 9$ <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>≥ 4</td></tr> <tr> <td>O_i</td><td>17</td><td>31</td><td>19</td><td>14</td><td>19</td></tr> <tr> <td>E_i</td><td>12.2</td><td>27.0</td><td>28.5</td><td>19.0</td><td>13.3</td></tr> <tr> <td>$\frac{(O-E)^2}{E}$</td><td>1.89</td><td>0.59</td><td>3.17</td><td>1.32</td><td>2.44</td></tr> </table> $\sum \frac{(O-E)^2}{E} = \text{AWRT } 9.4$	x	0	1	2	3	≥ 4	O_i	17	31	19	14	19	E_i	12.2	27.0	28.5	19.0	13.3	$\frac{(O-E)^2}{E}$	1.89	0.59	3.17	1.32	2.44	$r = 28.5, s = \text{AWRT } 9$ Pooling M1 M1A1c.a.o. B1ft, B1ft both B1 A1cao (7)
x	0	1	2	3	≥ 4																					
O_i	17	31	19	14	19																					
E_i	12.2	27.0	28.5	19.0	13.3																					
$\frac{(O-E)^2}{E}$	1.89	0.59	3.17	1.32	2.44																					
(d)	defective items do <u>not</u> occur <u>independently</u> <u>or</u> <u>not</u> with <u>constant probability</u>	B1ft (1)																								
		13																								
(a)	M1 for attempt to find mean or $\hat{p}$ (as printed or better). The 0.1 must be seen in part (a).																									
(b)	M1 for correct expression for r or s using the binomial distribution. Follow through their $\hat{p}$.																									
(c)	1 st M1 for some pooling (accept $x \geq 5$, obs.freq. ... 14, 9, 10 and exp.freq. 19.0, s , 4.3) 2 nd M1 for calculation of test statistic (N.B. $x \geq 5$ gives 14.5). One correct term seen. 1 st B1ft for number of classes - 2 (N.B. $x \geq 5$ will have $6 - 2 = 4$) 2 nd B1ft for the appropriate tables value, ft their degrees of freedom. (NB $\chi^2_4(5\%) = 9.488$) 3 rd B1 (for hypotheses) allow just " $X \sim B(20, 0.1)$ " for null etc. 2 nd A1 for correctly rejecting Binomial model. No ft and depends on 2 nd M1.																									
(d)	B1ft for independence or constant probability – must mention defective items or defectives Follow through their conclusion in (c). So if they do not reject they may say "defectives occur with probability 0.1". Stating the value implies constant probability.																									

Q4.

The number of goals scored by a football team is recorded for 100 games. The results are summarised in Table 1 below.

(a) Calculate the mean number of goals scored per game.

(2)

Number of goals	Frequency
0	40
1	33
2	14
3	8
4	5

Table 1

The manager claimed that the number of goals scored per match follows a Poisson distribution. He used the answer in part (a) to calculate the expected frequencies given in Table 2.

(b) Find the value of r and the value of s giving your answers to 3 decimal places.

(3)

Number of goals	Expected Frequency
0	34.994
1	r
2	s
3	6.752
≥ 4	2.221

Table 2

(c) Stating your hypotheses clearly, use a 5% level of significance to test the manager's claim.

(7)

(Total 12 marks)

Parts (a) and (b) were answered very well and most scored full marks on these two parts but part (c) proved more challenging. Many insisted on including the mean of 1.05 in their hypotheses even though this was incompatible with their correct treatment of the degrees of freedom. The pooling of the last two groups was usually carried out and the calculation of the test statistic was often correct. There was some confusion over the calculation of the degrees of freedom though: many subtracted 2 but others only 1 and some were not sure whether to subtract from the number of classes before or after the pooling. A number failed to score the final mark because their conclusion was not given in context: comments such as "there is evidence to support the manager's claim" or "there is evidence that the number of goals scored in football matches does follow a Poisson distribution" are fine; "the data follows a Poisson distribution" is not.



Question Number	Scheme	Marks																		
(a)	$\lambda = \frac{0 \times 40 + 1 \times 33 + 2 \times 14 + 3 \times 8 + 4 \times 5}{100} = 1.05$	M1 A1 (2)																		
(b)	Using Expected frequency = $100 \times P(X=x) = 100 \times \frac{e^{-1.05} 1.05^x}{x!}$ gives $r = 36.743$ $s = 19.290$	M1 A1 A1 (3)																		
(c)	H_0 : Poisson distribution is a suitable model H_1 : Poisson distribution is not a suitable model <table border="1"> <thead> <tr> <th>Number of goals</th><th>Frequency</th><th>Expected frequency</th></tr> </thead> <tbody> <tr> <td>0</td><td>40</td><td>34.994</td></tr> <tr> <td>1</td><td>33</td><td>36.743</td></tr> <tr> <td>2</td><td>14</td><td>19.290</td></tr> <tr> <td>3</td><td>8</td><td>6.752</td></tr> <tr> <td>≥ 4</td><td>5</td><td>2.221</td></tr> </tbody> </table> 8.972443 $\nu = 4 - 1 - 1 = 2$ CR: $\chi^2_2(0.05) > 5.991$ $\sum \frac{(O-E)^2}{E} = \frac{(40-34.9937)^2}{34.9937} + \dots + \frac{(13-8.972443)^2}{8.972443}$ $= 4.356$ (ans in range 4.2 – 4.4) Not in critical region Number of goals scored can follow a Poisson distribution / managers claim is justified	Number of goals	Frequency	Expected frequency	0	40	34.994	1	33	36.743	2	14	19.290	3	8	6.752	≥ 4	5	2.221	B1 M1 B1ft B1 M1 A1 A1 ft (7) [12]
Number of goals	Frequency	Expected frequency																		
0	40	34.994																		
1	33	36.743																		
2	14	19.290																		
3	8	6.752																		
≥ 4	5	2.221																		
(a)	M1 for an attempt to find the mean- at least 2 terms on numerator seen Correct answer only will score both marks																			
(b)	M1 for use of correct formula (ft their mean). 1 st A1 for r , 2 nd A1 for s (19.29 OK)																			
(c)	1 st B1 Must have both hypotheses and mention Poisson at least once inclusion of their value for mean in hypotheses is B0 but condone in conclusion 1 st M1 for an attempt to pool ≥ 4 2 nd B1ft for $n-1-1 = 2$ i.e realising that they must subtract 2 from their n 3 rd B1 for 5.991 only 2 nd M1 for an attempt at the test statistic, at least 2 correct expressions/values (to 3sf) 1 st A1 for answers in the range 4.2~4.4 2 nd A1 for correct comment in context based on their test statistic and their cv that mentions goals or manager. Dependent on 2 nd M1 Condone mention of Po(1.05) in conclusion Score A0 for inconsistencies e.g. "significant" followed by "manager's claim is justified"																			